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Exp No: 12	**End-End Communication at Transport Layer - Echo Client Server and Chat**

AIM:

To implement (a) an echo client-server using TCP and UDP sockets and (b) a chat program using socket programming, for end-to-end communication at the transport layer.

ALGORITHM / PROCEDURE:

(a1) TCP echo server and client

1. **Server**: create a TCP socket with `socket(AF_INET, SOCK_STREAM)`, `bind()` it to an IP address and port (5000), call `listen()` and wait in `accept()` for the client.
2. **Client**: create a TCP socket and call `connect()` with the IP address and port of the server.
3. The client sends a message using `send()`. The server reads it with `recv()`, displays it and sends the same data back using `sendall()`.
4. The client receives the echoed message and displays it. This is repeated for each message.
5. Close the sockets using `close()`.

(a2) UDP echo server and client

1. **Server**: create a UDP socket with `SOCK_DGRAM` and `bind()` it to port 6000. No `listen()` / `accept()` is needed because UDP is connectionless.
2. **Client**: create a UDP socket and send the message to the server address with `sendto()`.
3. The server receives the datagram and the client address with `recvfrom()` and sends it back using `sendto()`.
4. The client receives the echo with `recvfrom()` and displays it. Close the sockets.

(b) Chat program

1. The chat server creates a TCP socket on port 7000 and waits for a client. The client connects to it.
2. The client types a message and sends it. The server displays it, then the server user types the reply and sends it back.
3. Messages are exchanged alternately until either side sends the word `exit`. Then both sockets are closed.

Point	**TCP**	**UDP**
Socket type	SOCK_STREAM	SOCK_DGRAM
Connection	Connection oriented (3-way handshake)	Connectionless
Reliability	Reliable, ordered delivery	No guarantee
Server calls	bind, listen, accept, recv, send	bind, recvfrom, sendto
Used in	HTTP, FTP, SMTP	DNS, DHCP, streaming

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DIAGRAM:

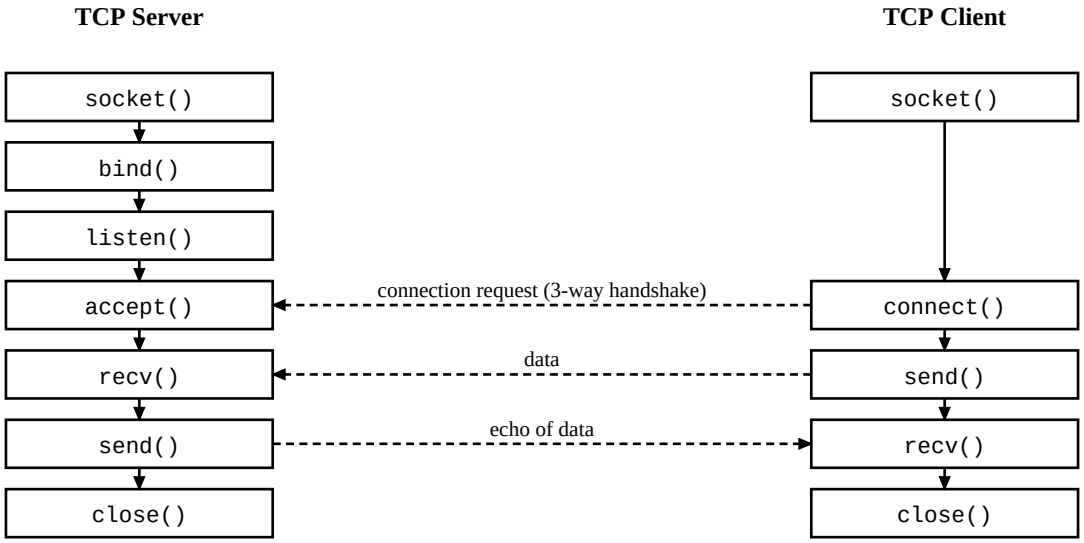

Fig 1: Socket calls in TCP client-server communication

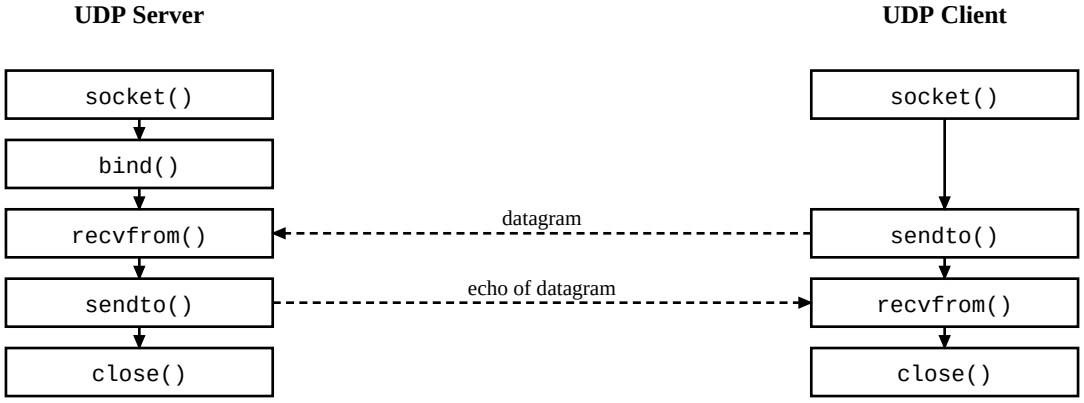

Fig 2: Socket calls in UDP client-server communication

PROGRAM:

echo_tcp_server.py

```
import socket
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.bind(("127.0.0.1", 5000))
s.listen(1)
print("TCP Echo Server listening on port 5000 ...")
conn, addr = s.accept()
print("Connected by", addr)
while True:
    data = conn.recv(1024)
    if not data:
        break
    print("Received :", data.decode())
    conn.sendall(data)               # echo back
conn.close()
s.close()
print("Connection closed.")
```

echo_tcp_client.py

```
import socket
c = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
c.connect(("127.0.0.1", 5000))
for msg in ["Hello Server", "Computer Networks Lab", "bye"]:
```

echo_tcp_client.py (contd.)

```
    c.sendall(msg.encode())
    print("Sent     :", msg)
    print("Echoed   :", c.recv(1024).decode())
c.close()
```

echo_udp_server.py

```
import socket
s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
s.bind(("127.0.0.1", 6000))
print("UDP Echo Server waiting on port 6000 ...")
for _ in range(2):
    data, addr = s.recvfrom(1024)
    print("Received from", addr, ":", data.decode())
    s.sendto(data, addr)
s.close()
```

echo_udp_client.py

```
import socket
c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
for msg in ["Hello UDP", "Datagram test"]:
    c.sendto(msg.encode(), ("127.0.0.1", 6000))
    data, _ = c.recvfrom(1024)
    print("Sent     :", msg)
    print("Echoed   :", data.decode())
c.close()
```

chat_server.py

```
import socket
s = socket.socket()
s.bind(("127.0.0.1", 7000))
s.listen(1)
print("Chat server started. Waiting for client ...")
conn, addr = s.accept()
print("Client connected:", addr)
while True:
    msg = conn.recv(1024).decode()
    if not msg or msg.lower() == "exit":
        break
    print("Client :", msg)
    reply = input("Server : ")
    conn.send(reply.encode())
    if reply.lower() == "exit":
        break
conn.close()
s.close()
print("Chat ended.")
```

chat_client.py

```
import socket
c = socket.socket()
c.connect(("127.0.0.1", 7000))
print("Connected to chat server. Type 'exit' to quit.")
while True:
    msg = input("Client : ")
    c.send(msg.encode())
    if msg.lower() == "exit":
        break
    reply = c.recv(1024).decode()
    if reply.lower() == "exit":
        break
    print("Server :", reply)
c.close()
print("Chat ended.")
```

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OUTPUT:

(a1) TCP echo

USER @USER -VirtualBox: ~ (Terminal 1 - Server)

```
USER   @USER   -VirtualBox:~$ python3 echo_tcp_server.py
TCP Echo Server listening on port 5000 ...
Connected by ('127.0.0.1', 46248)
Received : Hello Server
Received : Computer Networks Lab
Received : bye
Connection closed.
```

USER @USER -VirtualBox: ~ (Terminal 2 - Client)

```
USER   @USER   -VirtualBox:~$ python3 echo_tcp_client.py
Sent     : Hello Server
Echoed   : Hello Server
Sent     : Computer Networks Lab
Echoed   : Computer Networks Lab
Sent     : bye
Echoed   : bye
```

(a2) UDP echo

USER @USER -VirtualBox: ~ (Terminal 1 - Server)

```
USER   @USER   -VirtualBox:~$ python3 echo_udp_server.py
UDP Echo Server waiting on port 6000 ...
Received from ('127.0.0.1', 53205) : Hello UDP
Received from ('127.0.0.1', 53205) : Datagram test
```

USER @USER -VirtualBox: ~ (Terminal 2 - Client)

```
USER   @USER   -VirtualBox:~$ python3 echo_udp_client.py
Sent     : Hello UDP
Echoed   : Hello UDP
Sent     : Datagram test
Echoed   : Datagram test
```

(b) Chat program

USER @USER -VirtualBox: ~ (Terminal 1 - Chat Server)

```
USER   @USER   -VirtualBox:~$ python3 chat_server.py
Chat server started. Waiting for client ...
Client connected: ('127.0.0.1', 43260)
Client : Hi Server, this is USER
Server : Hello USER   , how can I help?
Client : Is the CN lab record ready?
Server : Yes, the lab record is due on Friday.
Client : Thanks, bye
Server : exit
Chat ended.
```

USER @USER -VirtualBox: ~ (Terminal 2 - Chat Client)

```
USER   @USER   -VirtualBox:~$ python3 chat_client.py
Connected to chat server. Type 'exit' to quit.
Client : Hi Server, this is USER
Server : Hello USER   , how can I help?
Client : Is the CN lab record ready?
Server : Yes, the lab record is due on Friday.
Client : Thanks, bye
Chat ended.
```

RESULT:

Thus the echo client-server programs using TCP and UDP sockets and the chat program using socket programming were implemented and executed successfully.